

PRANAY THARALA

Senior AI / ML Engineer

t.pranaytharala@gmail.com | (201) 268-9549 | linkedin.com/in/pranay-t-35157035b

PROFESSIONAL SUMMARY

Senior AI/ML Engineer with 7+ years of software engineering experience and 3+ years building production LLM and machine learning systems. Deep hands-on expertise across the GenAI stack: LLM application development, RAG pipelines, agentic AI systems with tool use and orchestration, and LoRA/QLoRA fine-tuning. Shipped AI products in regulated financial services (American Express, Vanguard) with a high bar for reliability, evaluation, and responsible AI. Strong Python backend engineer (FastAPI, REST APIs, microservices) across AWS, GCP, and Azure, with MLOps/LLMOps practices covering the full model lifecycle from experimentation to production monitoring.

TECHNICAL SKILLS

AI / GenAI: LLM application development, RAG pipelines, AI agents & orchestration, tool calling, prompt engineering, LoRA/QLoRA fine-tuning, LLM evaluation (TruLens, LLM-as-a-judge), latency / cost / accuracy benchmarking, NLP (entity extraction, classification, summarization), conversational AI, voice AI (Whisper)

Frameworks & Libraries: LangChain, LangSmith, Hugging Face Transformers, PyTorch, TensorFlow, Keras, JAX, scikit-learn, XGBoost, spaCy, NLTK, BERT

Models & LLM Platforms: Llama 3, Mistral, OpenAI API, AWS Bedrock, GCP Vertex AI, Azure AI

Data & Vector Search: Pinecone, FAISS, OpenSearch, pgvector, SQL, BigQuery, Airflow, data pipelines

Big Data & ML Platforms: Databricks, Apache Spark, Spark MLlib

Computer Vision: OpenCV, image/video analytics, OCR, object detection

Distributed Training & Compute: CUDA, Ray, distributed training

BI & Visualization: Tableau, Power BI

Backend Engineering: Python, Java, R, FastAPI, Spring Boot, REST APIs, microservices

MLOps / DevOps: MLflow, Kubeflow, Docker, Kubernetes, Terraform, Jenkins, GitLab CI, CI/CD, experiment tracking, automated retraining

Cloud & Infrastructure: AWS (SageMaker, EC2, S3, RDS, Lambda), GCP, Azure

Quality & Responsible AI: responsible-AI reviews, bias-mitigation controls, model-evaluation criteria, Grafana, Splunk

WORK EXPERIENCE

American Express | Senior AI Engineer

Mar 2026 - Present

- Design, develop, and deploy production GenAI applications — RAG pipelines, multi-agent systems, and LLM-powered APIs — for fraud detection and document intelligence.
- Build agentic AI systems (LangChain) with tool use, vector-database retrieval (Pinecone, FAISS), and event-driven orchestration that triage fraud alerts end to end.
- Develop LoRA/QLoRA fine-tuning strategies (Llama 3, Mistral) and prompt-engineering practices that reduce hallucinations in compliance-heavy workflows.
- Evaluate LLM systems with TruLens and LLM-as-a-judge; run systematic latency/cost/accuracy benchmarking across LLM providers (OpenAI API, AWS Bedrock, GCP Vertex AI).
- Apply XGBoost/gradient-boosting models with rule-based logic to automate anomaly detection, reducing false positives and manual investigation effort.
- Lead responsible-AI reviews, model-evaluation criteria, and bias-mitigation controls; mentor engineers on shipping LLM features safely.

Environment: Python, LangChain, RAG, LLM Agents, Pinecone, FAISS, TruLens, LoRA/QLoRA, XGBoost, Databricks, Apache Spark, OpenCV, Tableau, Ray, AWS, GCP Vertex AI

Vanguard | AI/ML Engineer

Sep 2023 - Mar 2026

- Architected and deployed RAG pipelines (LangChain, Pinecone, FAISS, Vertex AI; evaluated OpenSearch, pgvector) over fund/account documentation, delivering cited, context-aware answers via APIs.
- Built LLM-powered agents and event-driven APIs (Java, Spring Boot, AWS Lambda) bridging AI platforms with enterprise data systems.
- Designed LoRA/QLoRA fine-tuning strategies for Llama and Mistral on domain datasets; integrated multiple LLM APIs behind a common interface with systematic benchmarking.
- Implemented MLOps/LLMOps (MLflow, Airflow) for experiment tracking, versioning, and automated retraining; built BERT/spaCy NLP pipelines for entity extraction and classification.
- Developed XGBoost models for risk scoring and prediction.

Environment: Python, Java, Spring Boot, LangChain, Pinecone, FAISS, AWS Lambda, GCP Vertex AI, LoRA/QLoRA, XGBoost, Spark MLlib, Keras, NLTK, Power BI, MLflow, Airflow

HCLTech | Software Engineer

Jul 2020 - Jul 2023

- Designed and maintained backend services and RESTful APIs using Java and Spring Boot for global client applications.
- Deployed cloud infrastructure on AWS (EC2, S3, RDS, Lambda) and Azure, automating provisioning with Terraform.
- Built CI/CD pipelines (Jenkins, GitLab CI, Docker, Kubernetes), shortening release cycles and cutting manual deployment effort by over 30%.
- Wrote and optimized SQL across Oracle, PostgreSQL, and MySQL; implemented monitoring with Grafana and Splunk.

Environment: Java, Spring Boot, AWS, Azure, Terraform, Docker, Kubernetes, SQL, Jenkins, GitLab CI

HCLTech | Technology Trainee — TechBee Apprenticeship

Jul 2019 - Jul 2020

- Completed work-integrated apprenticeship covering Java, Python, data structures, RDBMS/SQL, and SDLC; converted to full-time Software Engineer.

PROJECTS

Fraud-Detection Virtual Assistant: Agentic AI assistant with vector-database retrieval triaging fraud alerts end to end, cutting analyst investigation time.

Enterprise Document-Intelligence RAG Assistant: Retrieval-augmented Q&A over fund/account documents (LangChain, Pinecone, FAISS, Vertex AI) with cited answers for compliance teams.

Fine-Tuned Financial-Domain LLM: LoRA/QLoRA fine-tuning of Llama and Mistral on compliance/transaction datasets, reducing hallucinations and improving domain accuracy.

EDUCATION

- Master's Degree, Information Technology — St. Francis College
- Bachelor of Computer Applications (BCA) — SASTRA University